Optimization of Movement: A Dynamical Systems Approach to Movement Systems as Emergent Phenomena

Andrew A. Guccione, Brian T. Neville, Steven Z. George

For over 30 years, the physical therapy profession has openly pursued the concept of diagnosis.[1,2] More recently, the attention on diagnosis has shifted to postulation of a "movement system" as the unique and overarching construct of the profession's body of knowledge and the foundation of physical therapist practice, as such practice seeks to optimize movement because movement is related to health.[3,4] The term "movement system" has been defined as "represent(ing) the collection of systems (cardiovascular, pulmonary, endocrine, integumentary, nervous, and musculoskeletal) that interact to move the body or its component parts."[5] Although we do not dispute the advantage of defining the "movement system" as a physiological system, we contend that how the profession identifies with a monolithic "movement system" is imprudent.

The "movement system" is characterized as a system of interacting systems that function in an *unspecified* way to move the human body, a nearly tautologous and instrumental definition of itself (ie, "movement system is" as the "movement system does"). In the current conceptualization of this system, what structure or system of the human body does the "movement system" exclude? What is gained by constructing an alternative and unfamiliar label to encompass all the systems of the human body? What does such a label add, scientifically and clinically, to the ability of physical therapists to deliver value-based care to patients? Although we concede a degree of professional political usefulness in labeling a system with which to identify our profession, the emphasis on reframing our identity that binds the profession to "the movement system" misses a critical opportunity that is unfolding to define and communicate professional expertise.

Diagnosis at the Crossroads

The case for diagnosis by physical therapists has been propelled by 2 assumptions. The first, and to some proponents the primary, assumption was that an autonomous profession must have a unique set of labels to apply to a patient's clinical presentation that could distinctly mark the domain of the profession,[2] and describe specific clinical presentations that would be broadly recognized by society as within the purview of the physical therapist.[6] Furthermore, these labels had to be framed in a way that would indicate that the clinical presentation was amenable to physical therapist intervention, and phrased in a way to direct treatment.[6] As the bulk of physical therapist interventions were treatments to ameliorate impairments, an impairment-based approach to diagnosis would presumably align labels with treatment procedures well within the purview of the profession.

Concomitantly, the second premise for developing diagnosis in the profession purported that physical therapist practice could not advance as a clinical science unless the diagnostic system used by physical therapists classified specific clusters of patient signs and symptoms in a way that would allow identification of nonoverlapping patient groups that could be used to establish treatment efficacy and make comparisons across patient types.[7,8] Yet, earnest endeavors by hundreds of physical therapist clinicians and scientists laboring to develop an impairment-based diagnostic classification system over 30 years have yielded little fruit. However, to date, few attempts have been made to identify whether the quest for diagnosis was misdirected, or to reassess what problem was to be solved in the process.

The current impasse in developing diagnosis in attempting to define the profession's expertise is partly due to a long-standing concern that physical therapy should be defined as a profession by what its practitioners know and not what they do. Anxiety over professional identity could have had unintended consequences that conflated the 2 driving assumptions on the need for diagnostic labels: firstly, the need to name what is to be treated, and secondly, the need to have a *unique* set of diagnostic labels. Furthermore, there has been a tendency to regard the diagnostic label, unique or not, as unchanging over the course of treatment. However, is it possible that the profession has misunderstood its concern? Maybe knowing what is to be done is a form of knowledge that is not reducible to the procedures or tools (eg, therapeutic exercise, manipulation, functional electrical stimulation) used in implementing that knowledge. Physical therapists possess a unique perspective, drawn from multiple knowledge bases, on interventions to optimize movement after injury or illness. Moreover, has the need to name what physical therapists treat blindsided the profession's understanding of the dynamic nature of continuing the diagnostic process after initial treatment has been implemented? We contend that it is time to move the profession's scientific expertise regarding "movement optimization" forward by reconsidering how movement is produced (and potentially optimized) as a behavioral

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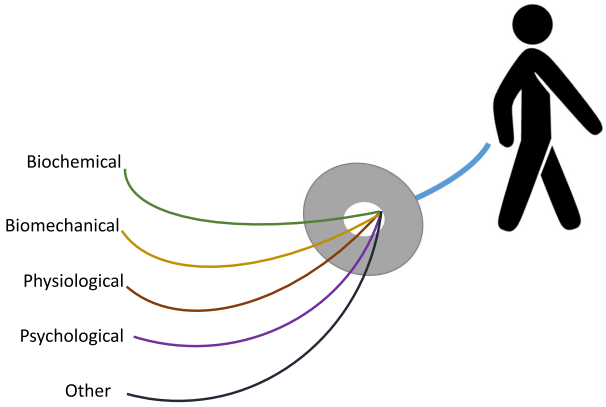


Figure 1.
Integration of multiple, nonlinear inputs (ie, capabilities/resources, constraints) from which purposeful movement emerges.

phenomenon in itself, and abandon further attempts to promote "the movement system" with a purportedly unique and static label.

Movement as an Emergent Phenomenon

The primary purpose of all rehabilitation efforts is to enable human functional performance to accomplish personally desired tasks and activities under real-world conditions. As a concept for physical therapist practice, purposeful movement can be framed as goal-directed behavior that allows an individual to maintain a posture, transition to other postures, or sustain actions until a desired end point is reached. Acknowledging movement as a primary mediator of human performance in this way supports the centrality of movement to the discipline's body of knowledge.

Purposeful movement requires a complex integration of capabilities and abilities of the individual (Fig. 1). Furthermore, we note that the profession has built a substantial evidence base to support examination and intervention when any of these capabilities or abilities is impaired, either alone or in concert with each other. In fact, virtually all the impairment-based approaches to intervention that dominate the history of the profession are dependent on identification of impaired body structures and functions that preclude, interrupt, or otherwise impede movement.[9] In the past, such intellectual commitments have pushed the profession toward diagnostic classification schemata formed exclusively around impairment clusters to describe various failure points in this complex integration of systems.

In contrast to understanding impairment clusters separated out from the putative object of inquiry, it has been proposed that any movement can only be understood in context (ie, the specific task to be achieved and the environment in which the actions exhibited by the individual to achieve the task and accomplish the activity are performed).[10,11] Thus, movement in this view is not a dysfunctional aberration of ideal movement, but an attempt at a "solution," inextricably embedded in a particular set of conditions, directed toward resolving a practical problem at hand (eg, rolling, standing, walking, running, reaching, or grasping). Moreover, as these solutions are practiced and refined across an array of contexts, movement evolves into a generalized behavior. Thus, the correct name for what is to be treated (ie, the "movement system" diagnosis at any given point in time) must be viewed dynamically as a phenotypic behavioral expression and not as a static label. This is true even when the diagnosis is applied as an accurate description of the initial condition of the patient.

Constraints and Dynamical Systems

Kugler and Turvey12 argued that descriptions of goal-directed movement behavior should focus on the functions of an action, not on the specific anatomical units involved because "an act is functionally specific, its variability is in reference to preserving the function it fulfils rather than preserving any particular aggregation of body parts that it happens to involve."[12] Their decades-old contention is consistent with more recent theories that structurally different elements might perform the same function or yield the same output.[13] In order to extend this insight to observed dysfunction in movement, we must turn to the role of constraints in a dynamical systems model. Dynamical systems theory is a multidisciplinary approach to characterize how constantly changing, interdependent systems evolve over time.[14] Importantly, the application of a dynamic systems approach emphasizes that natural phenomena operate under principles that hold for multiple levels of analysis.[14] Constraints can be defined as any variable that regulates change in the behavior of the entire system.[11] Constraints do not dictate deterministic pathways to how a movement will be achieved, but rather restrict the *possibilities* available for configuring a movement.[15] Newell described 3 kinds of constraints that impose boundaries on movement: (1) organismic constraints arising within the individual at the level of structural and functional resources and limitations; (2) task constraints that are specific to the task to be performed; and (3) environmental constraints that are externally imposed by the temporospatial conditions surrounding the performance.[11] Based within dynamical systems theory, the organism, the task, and the environment all place constraints on each other such that movement emerges from the self-organizing resolution of these specific constraints to achieve the desired performance (Fig. 2).[15]

This approach to movement forces us to consider a radical departure from a narrow impairment-based approach to movement-related diagnoses in 2 ways. Specifically, the objective of the diagnostic process would be refocused to capture a performance deficit not as a failure of the organism, but as an attempt to effect an "acceptable"

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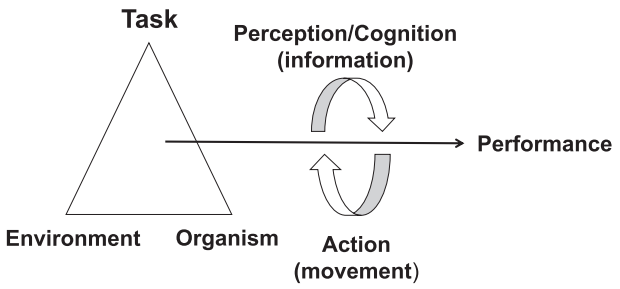


Figure 2.
Dynamical systems interpretation of resolution of the resources/constraints of organism, task, and environment emerging as a "movement solution" results in performance. (Adapted with permission of Springer Nature from Davids K, Glazier P, Araújo D, Bartlett R. Movement systems as dynamical systems: the functional role of variability and its implications for sports medicine. *Sports Med Auckl NZ*. 2003;33:245–260.)

solution out of the interactive constraints and resources among organism, task, and environment.[16] Examples of constraints implied or explicitly drawn from the literature that could be relevant to physical therapist practice beyond organismic constraints are found in the Table.[17–20] Furthermore, new or altered movements emerge when any of these control parameters in combination or sequence initiate a change in another control parameter. Moreover, these newly emergent movements may be inappropriately regarded as "the" movement dysfunction when considered only at the level of the organism. Thus, the object of the diagnostic process during initial examination should not be to arrive at a presumably static description of organismic impairment that can direct treatment over the therapeutic trajectory of an episode of care. In contrast, we propose an alternative approach to diagnostic labeling that interprets examination findings only as a set of initial conditions, which are relevant to the observed movement and potentially amenable to intervention, and can help to frame the first steps of treatment, but are not expected to "explain" the gap between observed and desired movement behaviors in any linear or fixed way. Furthermore, because the spontaneous, self-organizing nature of the movement system is simultaneously occurring on multiple levels of organization (eg, intra- and intercellular, tissue, organ, and system)[21] and is interdependent and dynamic, individual constraints and resources cannot be presumed to be static. Moreover, these initial diagnostic inferences about the confluence of constraints and resources that could be presumed to influence the observed behavior are used to generate hypotheses about an individual's performance, rather than using the data to place the individual categorically in a class of individuals with apparently similar clinical presentations.

Because every movement "solution" is individual, based on organismic, task, and environmental constraints, a singular "movement system" is only valuable if it is recognized as more than the sum of its parts, and that any emergent organism-level phenomenon (eg, movement, pain) is not equivalent to the system itself. Nor should the notion of a singular system convey an ideal system, either in form or in function. Rather, given that each individual self-organizes recovery by processes of compensation and adaptation, it would be more appropriate to emphasize multiple movement solutions in achieving a specific performance. Importantly, these movement solutions should be determined independent of the context of being in rehabilitation. We readily recognize that certain movement solutions might be "ideal" in achieving particular biomechanical outcomes (ie, reducing load on a particular anatomical structure in pain or at risk), but we should take care not to equate "ideal" movements under certain task and environmental constraints as optimal movement solutions to be applied across task and environmental conditions. Indeed, we contend that the concept of organism-centric "optimal movement" grounded in the patho-medical model should be avoided in favor of bounded variability that emerges from the interactions between organism, task, and environment (Fig. 3).

Bounded Variability

Fetters has summarized a compelling case for the critical contributions of variability to movement in infant and child development, which can be extended throughout the life span[22] and has already been recognized within sports medicine.[15] A different understanding of variability is needed when we accept the notion that although there are many ideal ways to move based on organismic constraints, optimal movement is dependent on task and environmental constraints concurrently. Therefore, variability in performance is not a consequence of error (ie, falling short of an "ideal" way to move) but rather an attempt at self-organization, potentially assisted by therapeutic interventions, to achieve a desired end performance within an acceptable range of error, at any given point in time. Neither too much nor too little variability in movement is necessarily desirable,[22] and variability may prevent inessential stability in complex environments that could compromise effective functional movement solutions in those environments. Indeed, Davids et al (2003) elegantly expresses how skillful performers are adept at exploiting the "high dimensionality" that our movement systems afford.[14] Evidence suggests that novices, in contrast, demonstrate weak task constraint adaptation due to difficulty with achieving successful pursuit of an outcome with acceptable bounds of variability.[15]

Optimization and Diagnosis

What then is the metric for judging optimization? There are many criteria that are specific to various clinical or research interests on which systems optimization might be quantified or qualified. Many existing tests and measures

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Table.
Constraint Constellations

Constraint Constellations—Organism	Examples of Issues to be Resolved	Does the Individual:
Structure	Integrity; alignment; load	Possess the structural support for the movement to be performed?
Ideation	Attention; intention; motivation	Understand the movement to be performed?
Initiation	Activation; recruitment; summation	Generate enough force to produce the desired movement?
Execution	Sequencing; timing; equivalence	Exhibit multisegmental actions without extraneous motions?
Equilibrium	Stability; transition; stopping	Stabilize after transitioning to a new posture?
Allostasis	Maintenance; adjustment; adaptation	Support appropriate thermoregulation and hydration?
Sustainability	Uptake; transport; extraction; utilization	Have the capacity to meet the energy demand of the activity?
Constraint Constellations—Task	**Examples of Issues to be Resolved**	**Does the Task:**
Equivalence	Novelty; transferability	Pose a familiar challenge?
Composition	Granularity; complexity; repetitiveness	Have multiple subcomponents?
Order	Sequence; flow; repetition	Need to be performed in a strict sequence?
Temporality	Speed; volume expectation	Impose a time limit?
Cognitive load	Intrinsic; extrinsic	Present only germane overload?
Bioenergetic demand	Capacity; intensity	Outstrip current capacity?
Proficiency	Skill level; tolerable variation	Have a minimum skill threshold?
Constraint Constellations—Environment	**Examples of Issues to be Resolved**	**Does the Environment:**
Location	Body stability; body transport	Fix on a particular place during task?
Predictability	Variability; familiarity	Pose unexpected challenges?
Context	Intensity; support	Promote attention and concentration?
Geography	Particularity; circumscription, terrain	Have strict boundaries of performance?
Regulatory control	Conformity	Impose restrictions on task execution?

of both impairments and treatment outcomes are still relevant to understanding the optimization of movement systems. When is a patient's performance as "good" as might be expected given interacting constraints at any particular moment in time? And when is today's "good" performance not good enough for tomorrow's expectations? As noted by Glazier and Davids:

> (O)ptimization may be better considered from the perspective of the individual under scrutiny and the confluence of constraints impinging on that individual, not some abstract, external reference or independent criterion. Since the constraints imposed on an individual dynamical movement system can fluctuate continuously over time, the optimal pattern of coordination and control for any given motor activity can change accordingly.[13]

Furthermore, Sparrow and Newell (1998) offer that "adaptive movement patterns emerge as a function of the organism's propensity to minimize metabolic energy expenditure with respect to task, environment, and organism constraints to action."[15] Thus, in this perspective, optimized movement successfully manages the perpetual balancing act of trade-offs between the work done in moving relative to its metabolic costs. Because movement efficiency is expressed as the quotient of a ratio, it becomes readily apparent that, mathematically, there are multiple relationships that could be considered equally efficient and therefore optimized relative to the constraints operative within a given situation. Although ratio comparisons of different trade-offs might appeal to the researcher, they pose particular challenges to clinical decisions.

Thus we would question diagnostic classification schemes that seek to group individuals on norms, are invariant across a class over time, and tend to discount the spectrum

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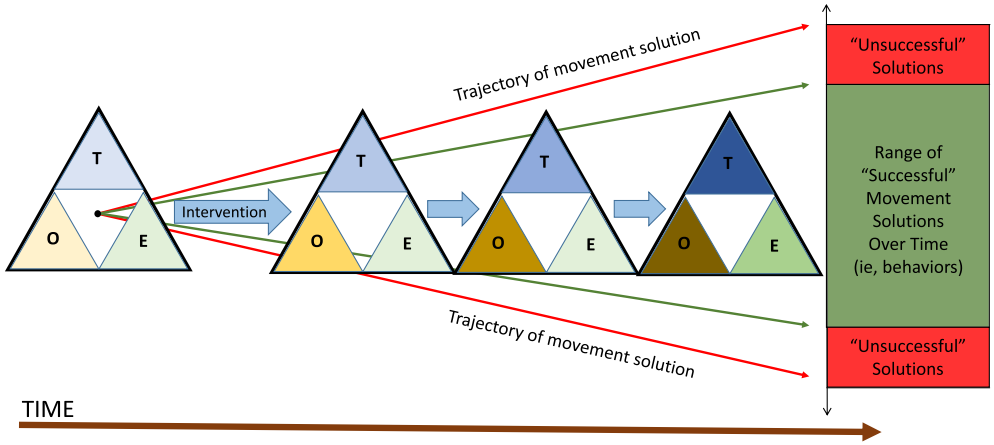


Figure 3.
Movement solution trajectories and resulting outcomes (ie, behaviors) as organismic resources and constraints adapt over time, including responses to intervention in the context of task and environment constraints. E = environment; O = organism; T = task.

of behaviors reflected in individuals' attempts to optimize a movement solution at any point in time. How can diagnosis of a system be approached given that constraints change over time, each change could be relevant, and salient features of the emergent movement are no longer dependent on initial conditions? Is it possible to establish a sensibility about diagnosis by physical therapists that recognizes the profession's indebtedness to other sciences, yet also captures our profession's unique history as interventionists in molding that broad scientific basis to address optimization of movement?

We believe that reframing our diagnosis is possible if there is a move away from an exclusive emphasis on classification of anatomical and physiological deviations from "normal" based on organismic constraints, when such data yield, at best, an incomplete insight into functional performance that includes environmental and task constraints. Instead, we propose pursuing criteria that emphasize the phenomena encompassed in human performance. We suggest that this can be accomplished through renewed focus on the interdependencies of the systems that comprise the human body, rather than on the structural or physiological systems themselves, as the basis for diagnosis by physical therapists.

Future Directions

Recently, Bittencourt et al advocated for transitioning to understanding sports injury etiologies through a "lens of complexity," a viewpoint that we believe is relevant to the current discussion.[21] Although the "black box" of movement systems to characterize the critical constraints might not be completely deciphered, physical therapists might do well to heed such suggestions to focus on interactions within what Philippe and Mansi dubbed a "complex web of determinants."[21,23] The examination procedures of physical therapists do not describe a complex and static integration of systems; instead those procedures describe a system integration that is often far from equilibrium, where minute changes result in nonlinear, disproportionate changes in the emergent phenomena.[24] Besides lacking the means to confidently parcel out only the critical parameters of the observed movement behavior at any particular time, such myopia ignores how complex systems self-organize across multiple dimensions, including time. Moreover, improving predictability in complex systems requires consideration of that system's own history, especially with respect to phenotypic response to treatment over time.

Future patient management schemata should continue to consider the traditional starting point for initial examination and evaluation (ie, various pathoanatomical and pathophysiological conditions), but with the caveat that these findings serve only as the initial conditions in identifying relevant patterns of interaction among organism, task, and activity over time. These initial conditions are essential to judging the potential for movement, but they cannot be considered in isolation.[25,26] To fully represent movement as the complex system that we intend to characterize, the initial pathoanatomical and pathophysiological conditions (answering the question "what is the underlying impairment?") must be considered as interactive parts of an emergent whole, leading to what is often a more pressing question for clinical decision-making (answering the question, "What is likely to happen?").[25,26]

The overall goal of having expertise in guiding movement systems toward optimal recovery should be to leverage adaptation by implementing therapeutic intervention in the context of that particular movement system's history of how the individual achieved a "solution" at that point in time. This aspirational goal suggests that initial diagnostic

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labels—if such labels are even necessary—would be multiaxial in order to capture 3 basic elements: (1) the fundamental initial conditions among the organism, the task, and the environment with respect to resources and constraints; (2) the potential targets for intervention to elicit intended adaptations over time; and (3) prognostication of the likelihood that the individual will achieve sufficient variability in the behavioral repertoire to achieve an outcome within the range of desirable potential outcomes. These diagnostic labels would only be used to represent initial system conditions; therefore, they need not be definitive or exhaustive. Consistent with systems science approaches, these labels should, in fact, be the bare minimum necessary to capture key parameters characterizing an individual's current performance, quantify and qualify the breadth and depth of the behavioral repertoire with respect to specific tasks and environments, and communicate the prognostic impact of findings, especially as the prognosis will affect patient management
strategies.

Interventions might then target these important relationships to demonstrate, through research and experience, outcomes relevant to successful performance rather than isolated remediation of impairments. In other words, the interventions would be based on the answer to the question "what is likely to happen over time?" and not their purported ability to counter the underlying impairment as the primary reason for implementing the intervention (Fig. 3). Although individual impairments may be ameliorated, the physical therapist would focus on establishing the conditions for restoring or improving relationships between the interdependent components of the movement system and finding resource and impairment combinations that were hypothesized as the most heavily "weighted"[21] toward contributing to successful emergent solutions to task accomplishment in the germane environment. Put another way, some constraints and resources will be more influential than others,[13] are necessarily entwined with the relationships to which they contribute, and might be more salient targets for intervention. Importantly, because humans are open, complex systems that never achieve a static equilibrium state[24] (ie, "normal"), restoration or improvement of individuals' function to ameliorate the performance gap is a continual process and, therefore, requires ongoing assessment of dynamic interdependencies. As such, clinicians' involvement is a dynamic process of creating the conditions for coaching adaptation.

The recent application of complex systems approaches to disciplines as diverse as medicine, biology, economics, and meteorology warrants thoughtful consideration of the potential benefits of incorporating similar advances in conceptualization of the central questions in physical therapy. The true potential of the profession's expertise in movement systems will be met when the examination is integrated into clinical decision-making primarily as a specification of initial conditions, with greater emphasis on key relationships that would enable improved prediction of an individual's future health and functional status, based on likely responses to given treatments over time.

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